

Spatial Patterns of Crime in King County, Washington (2019–2024): Hotspots, Trends, and Socioeconomic Drivers

Grace Harrill | GEOG 5790 Final Project

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2. Abstract

When looking at crime in cities it is not often distributed through communities evenly, instead it clusters around specific places such as intersections, streets, and specific neighborhoods. This project uses GIS based spatial analysis to understand the geography of reported crime in King County, Washington, focusing specifically on Seattle and surrounding neighborhoods between 2019 and 2024. The time period spans across six years of incident level data from the Seattle Police Department (SPD). Also incorporated with 2020 Census block groups, American Community Survey (ACS) socioeconomic variables, street centerlines, and neighborhood boundaries.

The analysis uses various spatial methods which include: annual Getis–Ord Gi* hotspot analysis to find statistically significant clusters, a hexagonal Space–Time Cube (STC) and Emerging Hot Spot Analysis (EHSA) to differentiate the crime over time, network-constrained Kernel Density Estimation (KDE) using street centerlines to find higher crime areas and Geographically Weighted Regression (GWR) to assess the relationships between crime rates and neighborhood characteristics and in the case of this study: renter households, poverty, and income. These variables were chosen because renter share, poverty, and income are key factors in shaping neighborhood crime patterns. All data preprocessing, spatial joins, and feature class generation were automated using ArcPy within ArcGIS Pro.

Results from this analysis show significant spatial concentration of crime. There were notable hotspots located in Downtown Seattle, Capitol Hill, and the University District, while coldspots are in more suburban communities such as Sammamish and Mercer Island. Network KDE was able to also indicate that there are linear crime areas along major roads like Aurora Avenue N and Rainier Avenue S. GWR implying that renter concentration is the most stable socioeconomic predictor of crime in this study, while poverty effects are more circumstantial dependent and income has minimal effect on crime rate based on current findings from this analysis.

3. Introduction

Crime that occurs in urban environments is inherently spatial. Criminological research has noted that crime concentrates at a small proportion of microplaces, often 3–5% of street segments or block groups accounting for 50% or more of incidents (Weisburd et al., 2016). These “hot spots” indicate a mix of land use patterns, and neighborhood social conditions, and they change over time as cities evolve as well. Understanding where crime clusters, how those clusters shift, and how they correlate to local socioeconomic conditions is useful for effective public safety policy and resource allocation.

This project focuses on crime patterns in King County, Washington, emphasizing the city of Seattle and its suburban surroundings. The study period, 2019–2024, covers three important periods: the pre-pandemic baseline, the COVID-19 pandemic and its disruptions to routine

activity patterns, and the post-pandemic era that has been distinguished by public concern over crime trends and, more recently, reports of significant declines (Seattle Times, 2025). Spatial analysis provides a way to move beyond citywide totals to ask: Which specific places experienced persistent crime problems, which improved, and which worsened?

This project uses a combination of incident level SPD crime data, Census block groups and ACS socioeconomic variables, TIGER road centerlines, and neighborhood geometry to explore crime in Seattle. The core emphasis is geospatial: preprocessing and analysis are carried out in ArcGIS Pro using ArcPy scripting, local spatial statistics, space–time modeling, and a network-based kernel density surface constrained to the street network. GWR is applied as a spatial model to illustrate how relationships between crime and neighborhood variables vary across space.

The goal is to provide a spatially direct interpretation of crime in King County that is congruent with both criminological assumption and current narratives about post-pandemic crime trends. Instead of exploring whether crime “went up or down” in Seattle, this project aims to ask where crime intensified, how those concentrations shifted from 2019 to 2024, and how neighborhood characteristics help describe these patterns.

4. Objectives

1. Map and indicate spatial clusters of crime in King County (2019–2024) using block group aggregation and Getis–Ord G_i^* hotspot analysis.
1. Analyze time series behavior of crime clustering using a hexagonal Space–Time Cube and Emerging Hot Spot Analysis to differentiate persistent, new, and diminishing hotspots.
2. Identify road based “crime corridors” using network-constrained Kernel Density Estimation (KDE) along street centerlines.
3. Look at spatially differing relationships between crime rates and neighborhood characteristics (renter households, poverty, and per-capita income) using Geographically Weighted Regression (GWR).
4. Compare urban core and suburban areas in regard to crime intensity, hotspot persistence, and model interpretability.

5. Methods

5.1 Study Area

The study area covers Seattle and its surrounding communities within King County, Washington (FIPS 53033). King County spans around 2,134 square miles and had a population of about 2.27

million in 2020. The region includes concentrated urban neighborhoods (for example, Downtown, Capitol Hill, University District), older suburbs (e.g., Renton, Kent), and higher-income suburban and exurban areas (e.g., Sammamish, Mercer Island, Shoreline).

In order to ensure accuracy all datasets used the following projection NAD 1983 HARN StatePlane Washington North FIPS 4601 (EPSG 2285). This projection was chosen to make all distance based measures (neighbor definitions for Gi*kernel bandwidths for KDE and GWR) in relevant linear measurements.

5.2 Data Sources and Description

Key datasets are summarized in the table below:

Dataset	Source	Geometry	Key Attributes	Use in Analysis
SPD Crime Data 2008- Present (filtered 2019-2024)	Seattle Police Department	Points	Offense type, timestamps, XY	Incident-level crime locations and attributes
2020 Census Block Groups (King County) with ACS 2019–2023	U.S. Census Bureau & ACS	Polygons	Population, renter households, families below poverty, per-capita income	Spatial units for aggregation and GWR predictors
TIGER/Line 2021 Roads (King County subset)	U.S. Census Bureau TIGER/Line	Lines	Street centerlines, addressing ranges	Street network for snapping incidents and network KDE

The SPD dataset includes granular incident-level information for all reported crimes, but this project only pertains to incidents between January 1, 2019 and December 31, 2024 to focus on a sex relevant time periods that surround the pandemic. Block-group geometries and ACS attributes were acquired via the University of Idaho GIS data portal and Census sources. TIGER road data were downloaded as county-level shapefiles and clipped to the block-group boundary.

5.3 ArcPy / GIS Programming Workflow

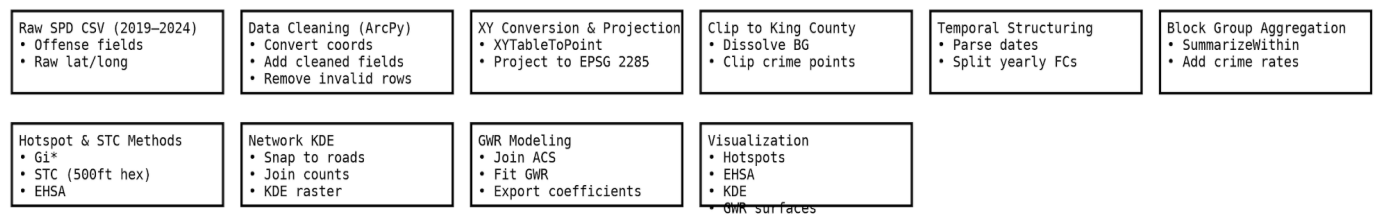
ArcPy was used to ensure consistent geospatial processing. The scripting workflow:

- Imported and cleaned raw SPD tables (coordinate parsing, date parsing).
- Converted cleaned XY coordinates to point feature classes in WGS84.
- Projected points into State Plane (EPSG 2285).
- Clipped points to the King County study area.

- Split incidents into yearly feature classes (spd_pts_2019–2024).
- Aggregated annual crimes to block groups using Summarize Within.
- Prepared inputs for Gi*, STC, EHSA, GWR, and network KDE.

Another script segment also incorporated using NumPy based export of block-group centroids and attributes for the mgwr GWR modeling, and another segment had road-based KDE inputs by snapping crime points to the street network and computing segment-level crime counts.

Below is my workflow visualized – *Read this workflow diagram from left to right; each box represents a sequential analysis step.*



5.4 Crime Data Pre-Processing

The raw SPD CSV was imported to a file geodatabase as **spd_raw**. Longitude and latitude were originally stored as text fields with formatting that was incompatible. Two new double fields (**Longitude_d**, **Latitude_d**) were created and populated, ensuring negative longitudes suitable for the Pacific Northwest.

Records were filtered to a bounding box (longitude -125° to -120° , latitude 47° – 48.6°) to remove obvious geocoding errors. The cleaned table was then converted to a point feature class (**spd_pts_wgs84**) using **XYTableToPoint**, then projected to State Plane North as **spd_pts_spn** using the **Project** tool. A Python function parsed the **Offense_Date** field to create an **OffDate_DT** date field and a **Year** integer field. To focus on King County, block group boundaries were dissolved into a single polygon and used to clip the crime points, producing **spd_2019_2024_spn**. Points were then separated into annual feature classes (**spd_pts_2019** through **spd_pts_2024**).

5.5 Aggregation to Block Groups

When working with different years the Summarize Within tool was utilized to ensure incidents were being located within their designated block group. Yearly layers were also created (**bg_2019_counts**–**bg_2024_counts**) with a **Join_Count** field. All block groups were kept, even those with zero incidents. For 2024, a crime rate per 1,000 residents (**Rate_2024**) was calculated using total population from the attributes table.

5.6 Hotspot Analysis (Getis–Ord Gi*)

The next part of this study was to find high or lower instances of crime cluster using Getis–Ord Gi* which is a local spatial statistic that is the comparison between the local sum of values within an assigned neighborhood to the global mean, taking into account spatial autocorrelation. High positive z-scores locate hotspots (high values surrounded by high values), high negative z-scores indicate coldspots. ArcGIS Pro’s *Hot Spot Analysis (Getis–Ord Gi)** tool was run on annual block-group counts ([Join_Count](#)) with these parameters: Conceptualization of spatial relationships: K-nearest neighbors ($k = 8$), distance method: Euclidean distance (in State Plane feet), and row-standardized weights.

Each year’s output ([hotspots_bg_2019](#)–[hotspots_bg_2024](#)) contained Gi_Bin classifications at 99% and 95% confidence levels, z-scores, and p-values. These outputs were mapped using a red to gray to blue ramp to illustrate patterns of noticeable clustering.

5.7 Space–Time Cube and Emerging Hot Spot Analysis

A **Space–Time Cube (STC)** was created using the [CreateSpaceTimeCube](#) tool. Crimes were compiled into hexagonal spatial bins with a 500-foot apothem and 6-month temporal intervals, yielding 12 time steps across the 2019–2024 period. The resulting NetCDF cube ([crime_STC_2019_2024.nc](#)) held counts for 103,728 space time bins, of which around 67% contained at least one incident.

A 3D scene was used to visualize the STC, with vertical extrusion representing incident counts. Spatial bins with 50 incidents were selected, showing approximately 360 “crime towers” (less than 2% of spatial bins) that contributed to a large share of total incidents.

The **Emerging Hot Spot Analysis (EHSA)** tool was run on the STC using a 1,000-foot neighborhood distance and a 1 time-step temporal lag. EHSA assigned each location to categories determined by Mann–Kendall trend statistics. This gave insight on how hotspot status shifted across the pandemic and post-pandemic years.

5.8 Network-Constrained Kernel Density Estimation (KDE)

To understand how transportation networks influence crime, a network-constrained KDE was used for this study. TIGER/Line 2021 roads for King County were projected to State Plane North and clipped to the block group boundary. To look at these networks over the course of time, yearly crime points ([spd_pts_2019](#)–[spd_pts_2024](#)) were snapped to the nearest road segment (within 30 feet) using the [Snap](#) tool. (All years were merged into a single feature class of snapped incidents). A spatial join calculated for each road segment, depicting the number of nearby incidents between 2019 and 2024. The Kernel Density tool was implemented to the center lines of roads using [Join_Count](#) as the population field, with a 10 unit cell size, and a 500

unit search radius. The resulting raster ([network_kde_2024.tif](#)) approximates relative crime intensity restricted to the specific street network. Values ranged from 0 (meaning no influence from nearby incidents) to about 132 (the highest modeled intensity).

5.9 Geographically Weighted Regression (GWR)

For 2024, a GWR model was created to explore spatially varying relationships between crime rate ([Rate_2024](#)) and these three predictors: Renter households (RENTER_HH_POPULATION), Families below poverty (FAMILIES_BELOW_POVERTY), and Per-capita income (PER_CAPITA_INCOME).

A spatial join added neighborhood names to [bg_2024_counts](#), and attributes were exported to NumPy arrays. The mgwr was a necessary Python package that was used with the kernel function, adaptive bisquare kernel and bandwidth selected by cross validation (Sel_BW), which in turn created 71 block groups, representing about 13% of all units. The model output created local coefficients and local R^2 values, which were put back into the block-group feature class and mapped to show where each predictor showed the most significance.

6. Results

6.1 Crime Rate Distribution

Crime rates in 2024 were highly distorted; this observation was made based on the fact that most suburban block groups had lower crime rates (often 20–100 incidents per 1,000 residents), while a small number of downtown and Capitol Hill block groups exceeded 1,000 incidents per 1,000 people. This map documents that a limited number of localized areas account for the crime that exists in the area.

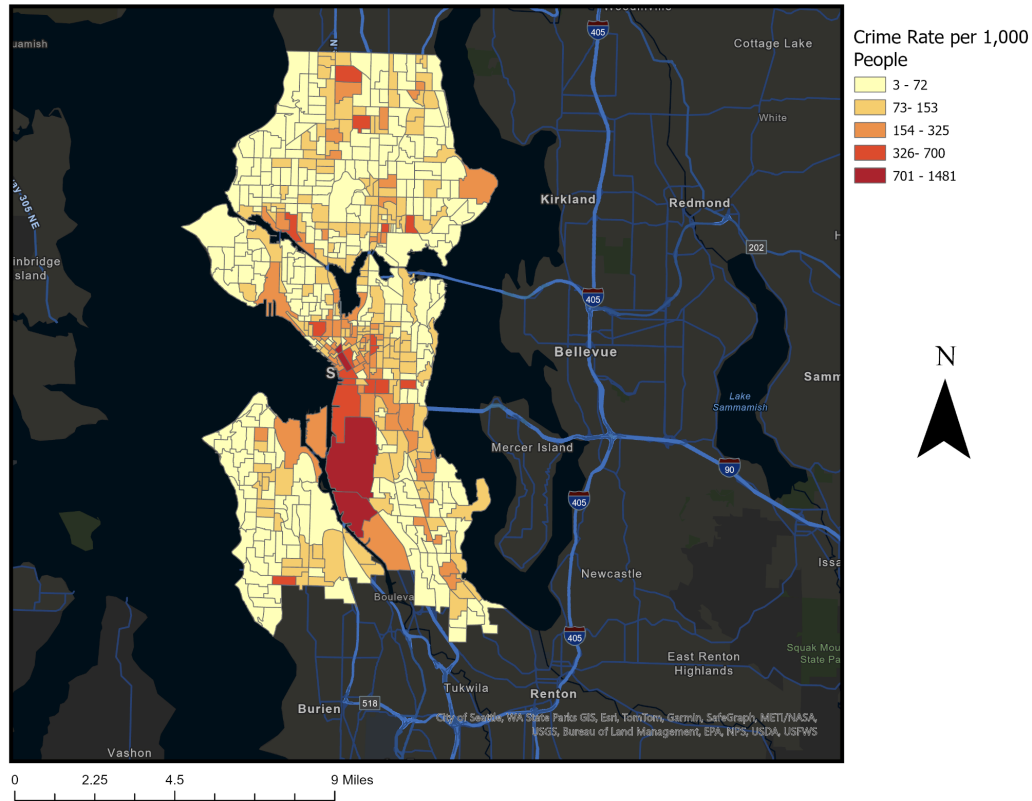


Figure 1. Crime Rate per 1,000 People (2024).
Block-group crime rates per 1,000 residents across Seattle and King County.

6.2 Annual Hotspot Patterns (Gi*)

Gi* maps for 2019 to 2024 showed that there were persistent hotspots in Downtown Seattle, Capitol Hill, and the University District. Coldspots appeared in high income suburbs such as Sammamish, Mercer Island, and certain parts of Shoreline. Also it is evident that areas around downtown and inner neighborhoods shifted between significant and less significant clustering. Between 2022 and 2024, the geographic extent of 99% confidence hotspots shrank noticeably, while non-significant areas expanded in neighborhoods going through social and economic changes.. This pattern aligns with reported citywide crime declines but provides more granular spatial detail.

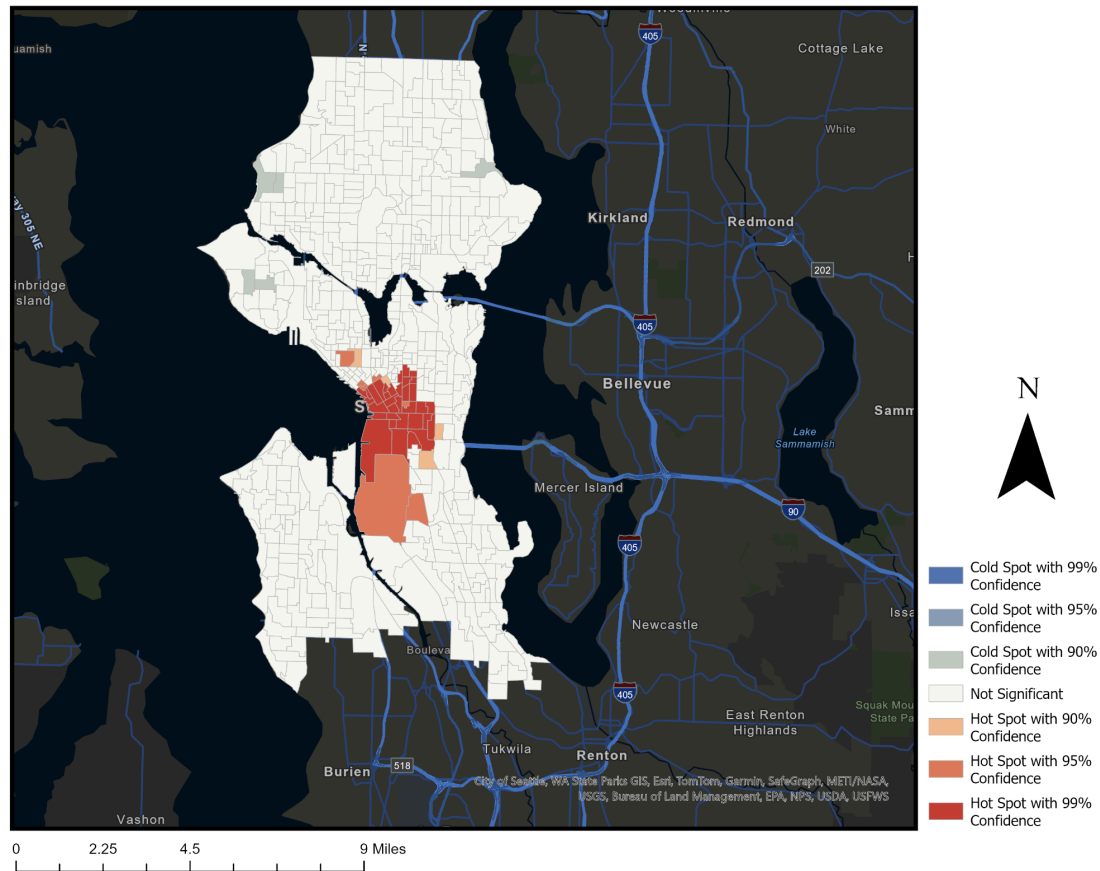


Figure 2. Gi* Hot Spot Confidence Mapping (2019–2024).

Getis-Ord Gi hotspots and coldspots by block group, symbolized by confidence level.*

6.3 Space–Time Cube and EHSA

The STC and EHSA emphasized the time based aspect of clustering. About 360 hexagonal “crime towers” (2% of spatial bins) showed high counts across multiple time steps, confirming significant spatial concentration. EHSA categorized much of Downtown, Capitol Hill, and U-District as persistent hotspots, with notable clustering across most of the time periods. New hotspots emerged in Northgate and parts of West Seattle in late-pandemic and post-pandemic years. Diminishing hotspots appeared in the Pike/Pine and Belltown areas after 2023, implying that some high crime entertainment districts experienced actual declines rather than citywide changes alone. Results show that the decline in crime is not evenly distributed but concentrated in particular corridors and districts within the county.

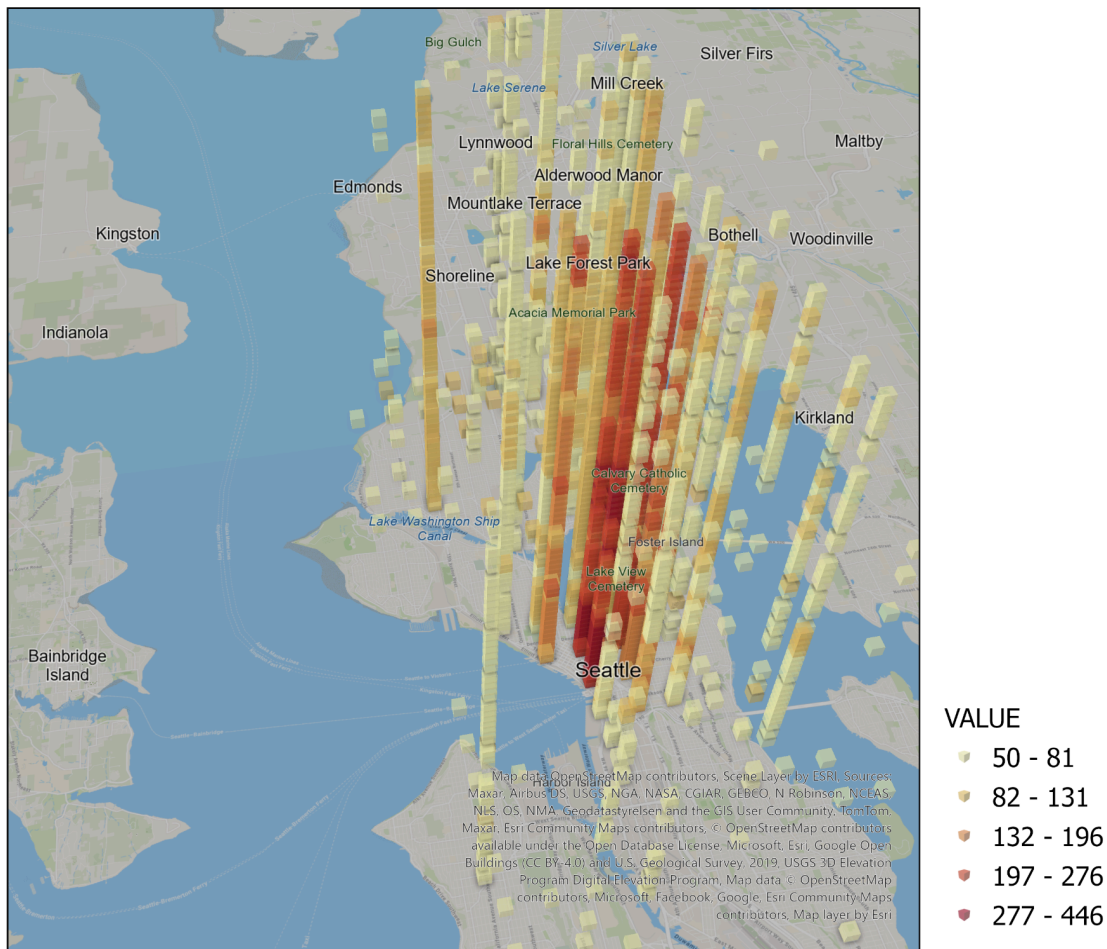


Figure 3. 3D Space–Time Cube Visualization.

The figure above is a Hexagonal space–time cube of crime counts, with taller red columns indicating persistent high-crime locations (Value in the legend corresponds to crime count)

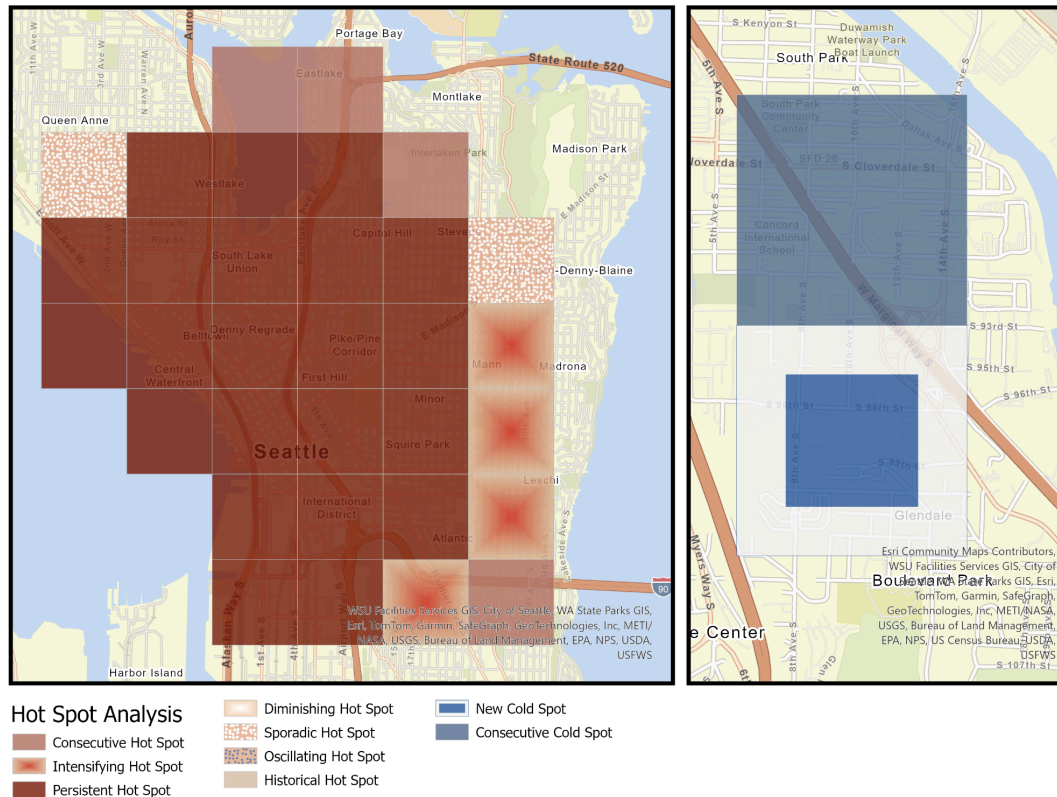


Figure 4. Emerging Hot Spot Analysis.
EHSA classifications showing persistent, new, diminishing, and cold spots for crime.

6.4 Network-Constrained KDE

The network KDE surface showed pronounced crime corridors, Aurora Avenue N / Highway 99 showed some of the highest density values, consistent with its role as a major arterial with motels, nightlife, and passing traffic. Downtown/Pike–Pine corridors and Belltown appeared as neighboring high-intensity segments. University District, Rainier Avenue S, and key arterials in West Seattle and Northgate also revealed raised densities.

The KDE surface also indicated that purely residential side streets and suburban areas had very low or no values. Because density was stilled to the street network, the outputs closely align with human mobility patterns and the physical structure of crime opportunities.

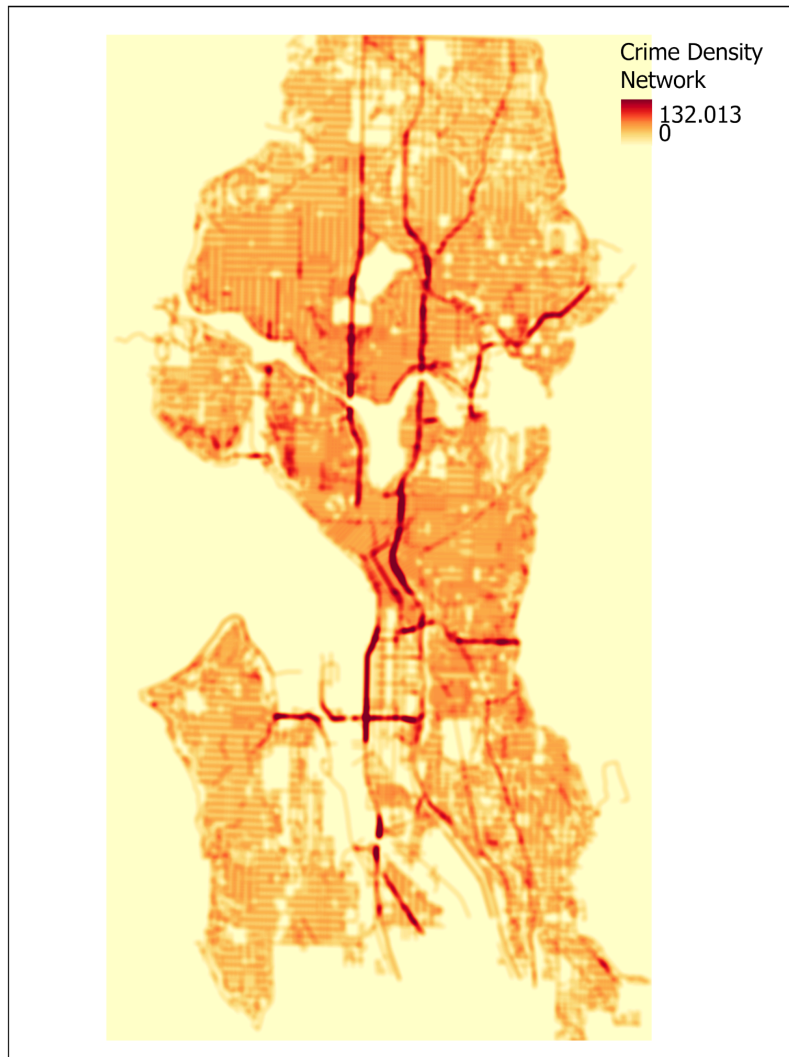


Figure 5. Network-Constrained Kernel Density Estimate (KDE)

Dark red segments indicate high-intensity crime locations—primarily major arterials, downtown corridors, and commercial strips—while pale yellow areas represent low or negligible density.

6.5 GWR Spatial Relationships

The GWR analysis showed spatial varying relationships in socioeconomic factors influencing crime throughout Seattle and King County. The resulting maps included the local R^2 surface, the renter coefficient, poverty and per-capita income surface.

Note: Coefficient surfaces for poverty and per-capita income are provided in Appendix C.

The analysis of local R^2 values showed values under 0.15 in entertainment and transit hubs, which is seen by non-resident activity, and exceeding 0.40–0.50 in suburban neighborhoods. This pattern indicates that socioeconomic structure is a strong predictor of crime when it comes to

residential areas but a poor predictor in commercial or nightlife areas. While the global OLS model explained only about 4% of variance, the GWR model showed a R^2 of around 0.27, emphasizing the importance of incorporating spatial diversity.

The renter coefficient was consistently positive across most block groups and strongest in higher turnover neighborhoods such as the University District, Belltown, and Capitol Hill. These patterns align with social disorganization theory: neighborhoods with high mobility and lower residential stability experience weakened social controls, supporting higher crime rates. Spatial variation in this effect highlights that renter concentration is a helpful predictor primarily in dense urban neighborhoods, while suburban areas show a more moderate but still positive association.

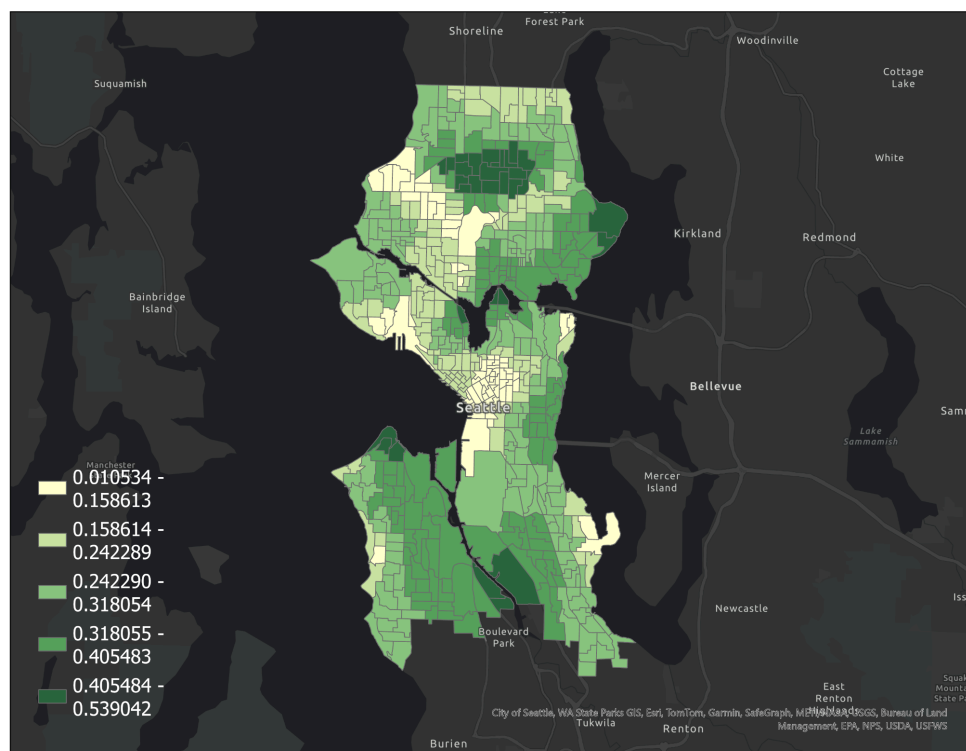


Figure 6. Model Local R^2 Surface (GWR).

Spatial variation in local R^2 values, indicating where the three-predictor GWR model fits crime rates well or poorly

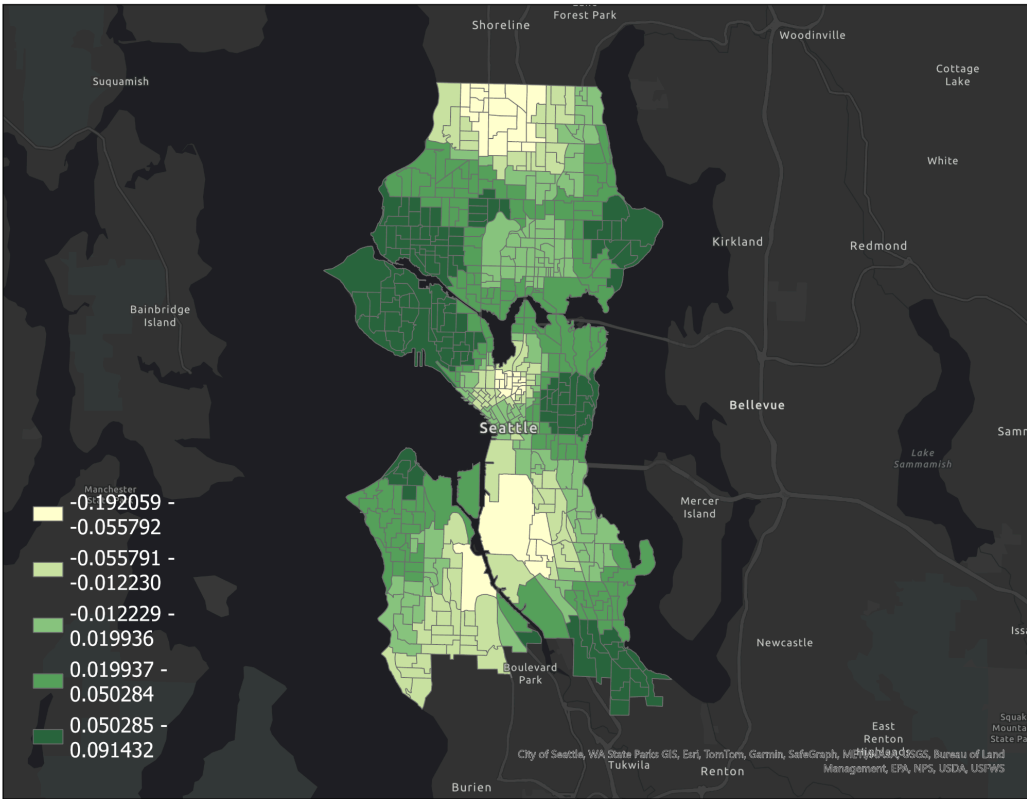


Figure 6. Renter Population Coefficient (GWR).

Local GWR coefficients for renter-occupied population, highlighting where renter concentration is most strongly linked to crime.

6.6 Hotspot Buffer and Supplemental Statistics

A 1,000-foot buffer around 95–99% Gi* hotspots showed:

- Crime rates 2.6 times higher inside buffers than outside.
- 62% more renter households and 38% more families below poverty in hotspot near block groups.

These results aid the claim that hotspots produce overflow effects in nearby areas, highlighting renter concentration as a crucial neighborhood risk factor.

7. Discussion

7.1 Spatially Uneven Crime Declines

These results aid in the ongoing discussion within the Seattle Times' (2025) reporting on recent crime declines but also show that these declines are spatially specific. Decreasing hotspots are concentrated in the Pike/Pine areas and Belltown, while recurring hotspots remain in other downtown and University District locations. Suburban areas were already low in crime and remained largely unchanged. This implies that interventions, economic changes, and shifts in routine activities have disproportionately impacted entertainment and nightlife districts, rather than producing uniform improvements across the city.

7.2 Socioeconomic Context and Spatial Heterogeneity

GWR results align with social disarrangement and localized areas theories. One being that renter concentration showcases residential instability, less ties socially speaking, and evidently lower investment in neighborhood institutions. High-renter neighborhoods consistently show stronger connections with crime, this is noticed more when there are also commercial uses involved. Poverty is subsequent but depends on the context at hand. In some neighborhoods, higher rates of poverty coincide with higher crime. While in areas, perhaps a strong community causes a buffer to higher crime. Income alone does not show any significant pattern when accompanied with crime data, as both high and low income areas can have their own unique and distinct support systems

7.3 Urban vs. Suburban Divergence

Dense urban areas such as Downtown, Capitol Hill, and the University District show higher crime rates and persistent hotspots. It is evident that local R^2 values are low in GWR, indicating that demographics alone cannot fully explain the variation. Instead, land use, nightlife, transit hubs, and non-resident populations have a more prominent influence.

Suburban communities are different in that they have lower crime rates and coldspots. There is also a higher local R^2 , indicating that socioeconomic variables do a better job explaining crime. For the most part residential crime types and more stable populations.

This difference between the two geographical areas suggests that policy responses should be tailored depending on where the neighborhood is located: environmental design and place based policing may be better suited in urban corridors, while housing stability and community development may be the solution in suburban areas.

8. Conclusions

Based on this study it can be concluded that a concentration like King County, shows spatially persistent hotspots, and social and economic factors cannot be the only variable that has an influence on why crime occurs in specific areas/neighborhoods. A small fraction of spatial units whether that be block groups, hexagonal bins, or road segments account for a disparate number of total incidents when looking at 2019 to 2024, aiding in the information that has been put forth by current crime analyses.

Disruptions that were caused by the pandemic did cause a shift in crime patterns, though not uniformly. The 2020–2021 period shows a quantifiable shift in the make up and intensity of hotspots. Some nightlife and commercial districts were observed as more quiet for a period of time due to business closures and there was reduced mobility. This is especially true in high density renter neighborhoods who saw comparatively stable or increased activity. The STC and EHSA results represent these short term dips followed by post 2022 rebounds, illustrating how sudden events like the pandemic reshape but do not remove fundamental spatial structures.

Hotspots are persistent but not static, this is evident in Downtown Seattle and Capitol Hill remain long term hotspots, but there are several entertainment districts that have declining intensity after 2023, while some outlying corridors show emerging or intensifying activity.

Crime follows the street network in Seattle, the network-constrained KDE highlights continuous crime corridors along arterials (Aurora Ave N, Rainier Ave S, Lake City Way) that standard polygon analyses did not capture. This supports routine activity theory: crime opportunities cluster along mobility pathways, not just within the given census boundaries.

Renter clustering is the strongest and from the chosen variables the most consistent predictor of elevated crime rates. Thus, displaying neighborhood turnover and less social cohesion. Poverty has localized effects meaning strongly positive in some South Seattle neighborhoods but weak or negative in more concentrated commercial areas where visitors, not residents, drive crime. Income alone performs poorly as a predictor once renters and poverty are included.

In Seattle urban areas tend to behave from suburban areas. Density, land use, and mobility have a direct effect on crime patterns downtown, though demographic characteristics better explain suburban imbalances. Spatial analysis frameworks such as STC/EHSA, KDE, and GWR enable researchers to incorporate theory with lived geographic patterns and to examine how extraordinary events like COVID-19 reshape, but do not entirely alter long standing spatial clustering of crime.

9. References

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Appendix B. Statistical Outputs

Table B. Block-Group Averages Inside vs. Outside 1,000-Foot Hotspot Buffer

This table is a comparison of the mean socioeconomic and crime rate values for block groups located within 1,000 feet of statistically significant 2024 Gi* hotspots versus those outside this zone. These statistics aid in the interpretation that hotspot areas show distinct socioeconomic characteristics.

Variable	Inside Buffer (Mean)	Outside Buffer (Mean)	Percent Difference
Crime Rate (2024)	191.45	74.38	+157%
Renter Household Population	911.04	563.86	+62%
Families Below Poverty	17.76	12.88	+38%

Block groups located near hotspots show significant higher crime rates and higher concentrations of renter households. This goes back to patterns identified in the GWR analysis. Poverty effects are present but less noticeable, aiding the finding that renter instability plays a stronger spatial role.

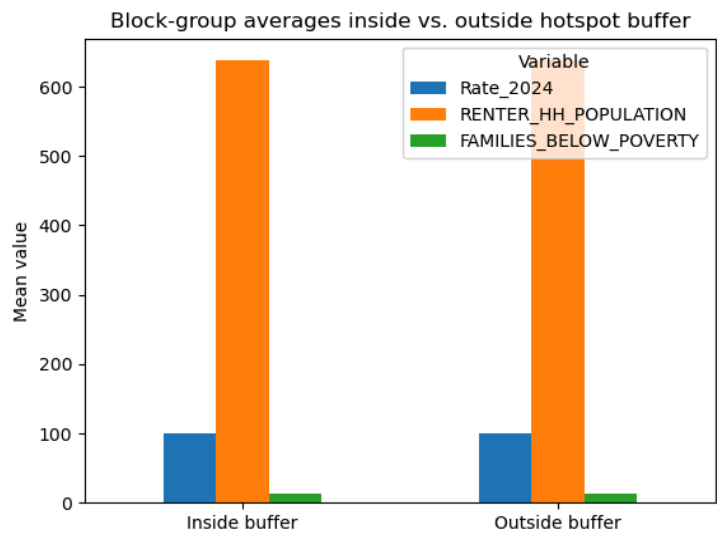


Figure B: Block-group Averages Inside vs. Outside Hotspot Buffer

The bar chart shows substantial differences between hotspot-adjacent areas and non-hotspot areas

A 1,000-foot buffer around 95–99% Gi* hotspots showed:

- Crime rates 2.6 times higher inside buffers than outside.
- 62% more renter households and 38% more families below poverty in hotspot near block groups.

These results support the notion that hotspots produce overflow effects in nearby areas, highlighting renter concentration as a crucial neighborhood risk factor.

Appendix C. GWR Coefficient Surfaces

This appendix provides all of the coefficient surfaces generated from the Geographically Weighted Regression (GWR) model described in Section 6.5. These supplementary coefficients for poverty and per-capita income are included here for transparency, and completeness of the analysis performed.

These maps visualize spatial non-stationarity in socioeconomic effects on crime and provide the more granular geographic detail. (See Figures C1 and C2 below)

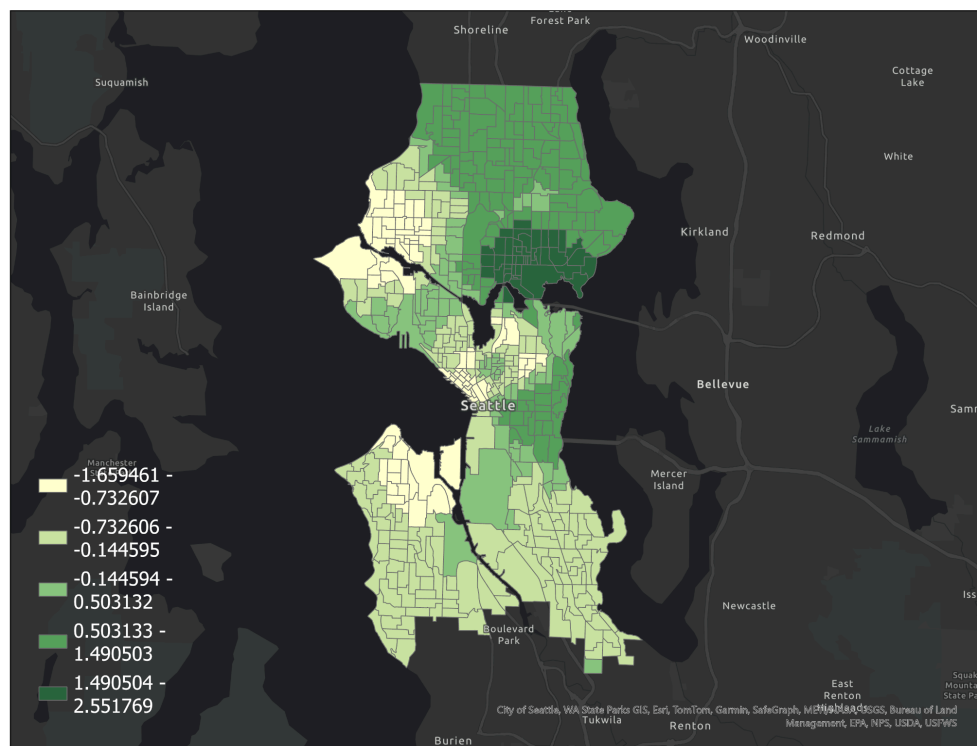


Figure C1. Poverty Coefficient (GWR)

Local GWR coefficients for families below the poverty line, showing where poverty is most strongly associated with crime.

This surface shows the spatial variation in the local effect of families below the poverty line on 2024 crime rates. Coefficients are different across the study area. Strongly positive values appear in South Seattle neighborhoods such as Rainier Valley and White Center, where poverty corresponds with historical disinvestment and weaker community resources. On the other hand, coefficients near zero or negative values occur in downtown and Capitol Hill, where crime appears more influenced by opportunity structures (nightlife, retail, non-resident traffic, etc.) than resident level socioeconomic disadvantage.

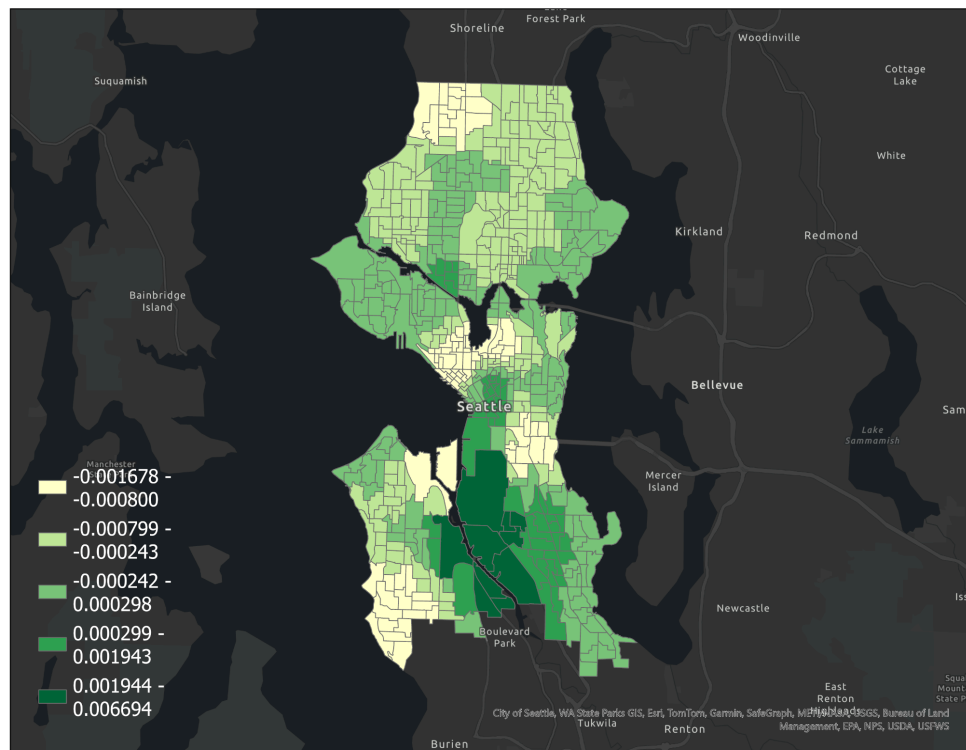


Figure C2. Per-Capita Income Coefficient (GWR).

Local GWR coefficients for per-capita income, showing the relatively weak and spatially inconsistent income effect.

This surface maps the local relationship between per-capita income and crime. Effects are small and there is a lack of spatial consistency, this corresponds with the weak global correlation. Some suburban areas show a slight negative association (lower income to slightly higher crime), These patterns suggest that income alone is an insufficient predictor of crime in Seattle.

Together, these supplemental figures demonstrate that poverty exhibits high spatial variability, indicating its effect on crime is strongly context-dependent. Income effects are weak, reinforcing

the conclusion that neighborhood mobility and instability (captured by renter concentration) play a more substantial role in shaping the spatial distribution of crime. These patterns validate the use of GWR instead of global regression, since no single coefficient value adequately describes the socioeconomic–crime relationship across King County.